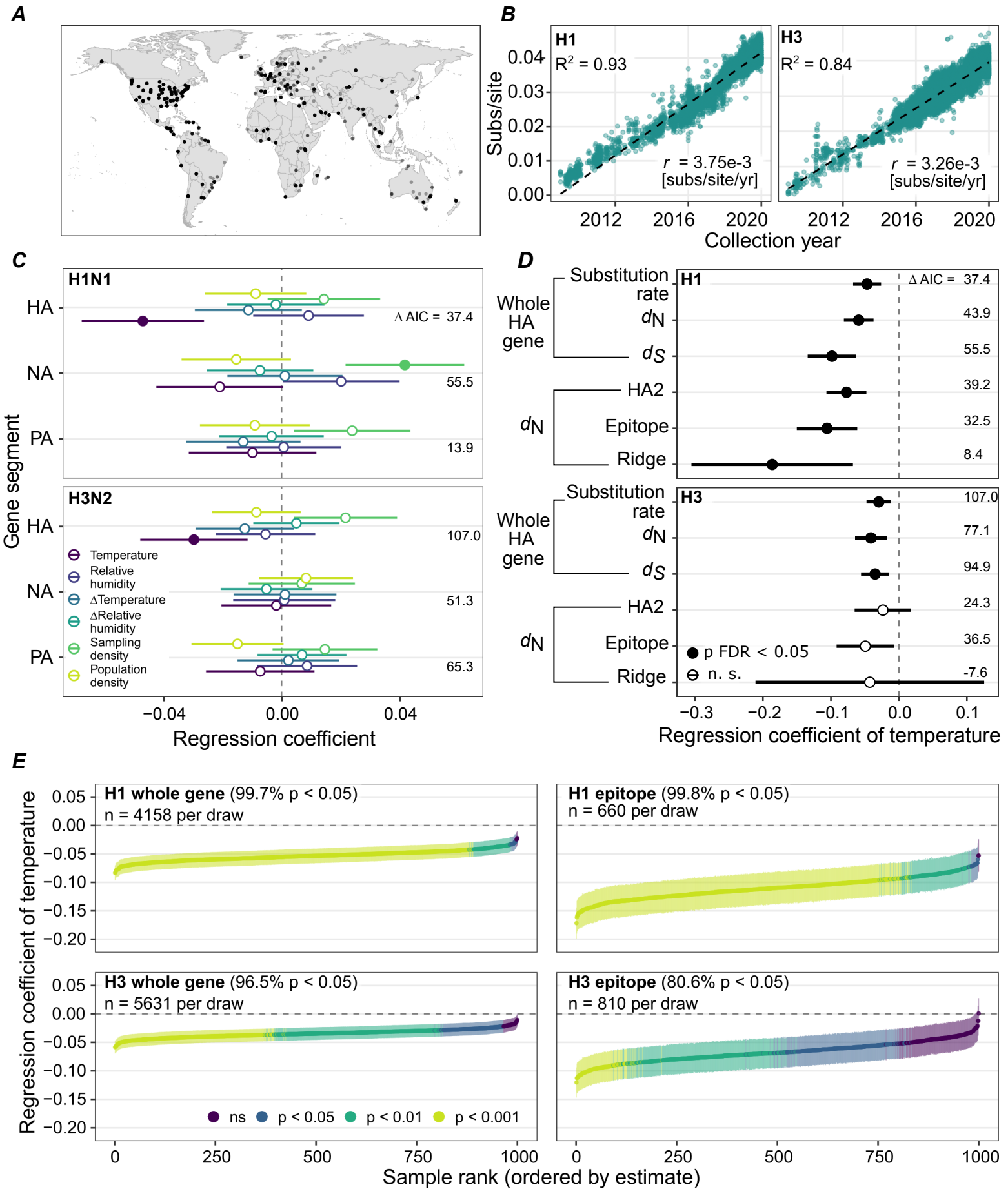

A
B
H1
R² = 0.93
r = 3.75e-3
[subs/site/yr]
H3
R² = 0.84
r = 3.26e-3
[subs/site/yr]
Subs/site
0.04
0.03
0.02
0.01
0.00
2012
2016
2020
Collection year
C
H1N1
HA
NA
PA
Δ AIC = 37.4
55.5
13.9
H3N2
107.0
51.3
65.3
Gene segment
Temperature
Relative humidity
ΔTemperature
ΔRelative humidity
Sampling density
Population density
−0.04
0.00
0.04
Regression coefficient
D
Substitution rate
Whole HA gene
dN
dS
HA2
Epitope
Ridge
H1
Δ AIC = 37.4
43.9
55.5
39.2
32.5
8.4
H3
107.0
77.1
94.9
24.3
36.5
-7.6
p FDR < 0.05
n. s.
−0.3
−0.2
−0.1
0.0
0.1
Regression coefficient of temperature
E
H1 whole gene (99.7% p < 0.05)
n = 4158 per draw
H1 epitope (99.8% p < 0.05)
n = 660 per draw
H3 whole gene (96.5% p < 0.05)
n = 5631 per draw
H3 epitope (80.6% p < 0.05)
n = 810 per draw
Regression coefficient of temperature
0.05
0.00
−0.05
−0.10
−0.15
−0.20
ns
p < 0.05
p < 0.01
p < 0.001
0
250
500
750
1000
Sample rank (ordered by estimate)